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Ultrasonic transducer probe holder

Abstract

An ultrasonic transducer probe holder suitable for holding an ultrasonic transducer probe of a medical ultrasound system to hold or secure the probe in place against a patient as an ultrasound technician releases the probe may include a grip portion and a skirt portion. The grip portion may be configured for deployment on the ultrasonic transducer probe. The grip portion may provide a comfortable and secure grip for the ultrasound technician as the technician applies the ultrasonic transducer probe against the skin of the patient. The ultrasonic transducer probe with the grip portion thereon may be insertable in the skirt portion of the ultrasonic transducer probe holder. The skirt portion may be configured to hold or support the ultrasonic transducer probe in an upward-standing position while forming a vacuum seal against the skin of the patient. A stand assembly may be configured for mounting on a bed, table, or other support on which the patient reclines. The stand assembly may be configured to hold or support the ultrasonic transducer probe holder as the holder remains deployed on the ultrasonic transducer probe. Accordingly, the stand assembly may maintain the ultrasonic transducer probe in place against the patient's skin to reduce or eliminate the manual pressure required for the technician to maintain the ultrasonic transducer probe in place during an echocardiography or other ultrasound procedure. The ultrasonic transducer probe holder may prevent carpal tunnel syndrome and other repetitive stress injuries in the hand of the technician.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. provisional application No. 63/322,259, filed Mar. 22, 2022, and entitled ULTRASONIC TRANSDUCER PROBE HOLDER, which provisional application is hereby incorporated by reference herein in its entirety.

FIELD OF THE INVENTION

(1) The present invention relates generally to medical ultrasound systems used to test for and diagnose heart and other medical conditions, and more particularly, to an ultrasonic transducer probe holder suitable for holding an ultrasonic transducer probe of a medical ultrasound system in place against the skin of a patient as an ultrasound technician manually releases the probe.

Background of the Invention

(2) Clinical medicine may utilize various procedures for diagnostic, therapeutic and/or monitoring purposes. These procedures may be carried out by a skilled operator. Examples of these types of medical procedures include diagnostic and therapeutic radiology, anesthesia, cardiology and surgery.

(3) Medical ultrasound procedures may be used extensively to image soft tissues as well as to carry out interventional procedures such as guidance or placement of needles or catheters. For example, ultrasound procedures may be used in the diagnostic imaging of organs such as the heart or liver. Interventional procedures that utilize ultrasound guidance include central line placement and guidance of nerve blocks. Ultrasound systems may be optimized for contrast and resolution in soft tissue.

(4) Ultrasound imaging is widely used in medical diagnostics procedures. Ultrasound imaging may provide high-resolution images of internal organs and biological tissues deep inside the body as well as functional information which indicates cardiac function and blood flow, for example. Ultrasound imaging may be carried out using an ultrasonic transducer. A typical ultrasonic transducer may include an array of elements that emit incident ultrasonic waves and receive reflected ultrasonic waves. The reflected ultrasonic waves indicate differences in acoustic impedance between respective organs and tissues and are converted into electric signals. Ultrasonic transducers may be capable of operation in a brightness mode, or B-mode, and a Doppler mode. In the B-mode, the ultrasonic transducer may use the received ultrasonic waves to generate two-dimensional images of the organ or structure in real-time. In the Doppler mode, the ultrasonic transducer may provide measurements including blood velocity. Doppler ultrasound may be used to estimate blood velocity by transmitting streams of high-frequency ultrasonic waves through the blood, receiving reflected ultrasonic waves reflected or echoed from red blood cells circulating in the blood and analyzing the reflected waves. A velocity profile of the blood is derived from measured changes in the phase of the reflected waves. Doppler scans may be used in the diagnosis of such conditions as heart valve defects, congenital heart disease, artery occlusions, and aneurysms.

(5) Early in a cardiac sonographers' training they are instructed on how to direct a narrow ultrasound beam between ribs of the chest wall and obtain orientations that allow for full interrogation of the heart chambers, vessels, and valves. Using a cigar sized probe in a pencil holding grip they perform a variety of rotations, twists and angulations focusing on obtaining the best quality images. Like a choreographed hand and arm ballet sonographers progress through the various views and obtain images in a coordinated standardized manner. These various contortions required in obtaining these images over time can be very taxing to the muscles and joints of the sonographer. In a 2009 sample by Evans et al, 90% of sonographers reported shoulder pain, with 69% reporting low back pain and more than half (54%) reporting work-related symptoms of the hand and wrist.² Unfortunately, there appears to be a culture of "toughness" which frowns upon individuals who might otherwise complain about their musculoskeletal discomforts. From generation-to-generation sonographers share stories of endurance and fighting through pains. Aches and pains have become the rites of passage of a seasoned sonographer. As a result of this work-related trauma international statistics indicate that 80% to 95% of sonographers experience work-related pain with 90% experiencing pain for more than half of their career and 1 in 5 sonographers sustain a career ending work-related injury.

(6) In typical implementation of an ultrasound procedure, an ultrasound technician, also known as a sonographer, may manually place a hand-held ultrasonic transducer against the patient. A water-based gel may be placed between the transducer and the skin of the patient. High frequencies of 7-18 MHz may be emitted from the transducer to image shallow structures, whereas lower frequencies of 1-6 MHz typically having better penetration may be used to image deeper structures. An image is generated and displayed in real time on a display. The sonographer may move the transducer to various locations or positions to obtain the desired image scan.

(7) Due to its manual nature, the output of an ultrasound procedure is highly dependent on the experience and capability of the technician performing the procedure. The results from an ultrasound procedure are not always reproducible between different technicians or scans. Furthermore, the manual process can be tiring for the technician and can cause musculoskeletal disorders due to sustained forces applied in unnatural positions. This may particularly be the case in applications in which the procedure is carried out on obese patients or patients with loose skin, in which it may be required that the transducer be pressed more firmly against the skin of the patient. Thus, a need exists for an ultrasound device that can automate the process of performing ultrasound imaging in order to achieve uniform, consistent, and reproducible ultrasound images.

(8) Work-related musculoskeletal disorders (WRMSDs) are painful injuries affecting the muscles, nerves, ligaments, and tendons of sonographers and of the users of diagnostic medical sonography. WRMSDs develop gradually over time from repeated exposure to risk factors and are among the most frequently reported causes of restricted or lost work time. The Bureau of Labor Statistics predicts an increase of 24% or 27,600 additional sonographers will be needed by 2024, an increase that exceeds average growth. With injury rates as high as 90%, it will be difficult to maintain a skilled work force. WRMSDs can impose a substantial personal toll on those affected since they may no longer be able to work or perform their simple personal tasks and activities of daily living.

(9) These workplace injuries are more prevalent because of intensified work schedules with less resting periods. New imaging modalities add to the already growing and protracted imaging protocols which is associated with increased imaging time and interaction between the sonographer and the computer. In general, the populous has become more obese with prevalence of lung disease both of which intensify the need of the sonographer to use more mechanical forces to improve the inherently poor image quality that these conditions cause. The force required in pushing, pulling, lifting, and gripping and pinching in addition to repetition of studies with very little recovery time and the inherent awkwardness of body position and alignment of head and neck shoulders and arm and wrist all intensify these work-related musculoskeletal traumas. The most common injuries among sonographers are carpal tunnel of the wrist, shoulder capsulitis and tendinitis, epicondylitis

of the elbow as well as neck and back strain.

(10) One simple intervention would consist of doing fewer studies and taking longer breaks however, employers typically incentivize doing more studies in the least time possible. Those individuals who might speak out and mention concerns of musculoskeletal trauma are generally viewed as slacker's and complainers. As a result, sonographers take their pain medications in secrecy making use of their hand and wrist supports and work until the pain subdues them into an early retirement or taking time off. Work schedule changes and ergonomic equipment design has only minimally mitigated the ultimate musculoskeletal trauma and possible career ending pain associated within the field of sonography.

(11) The industry standards for the prevention of Work-Related Musculoskeletal Disorders (WRMSDs) in the sonography is the work product of a 2016 Consensus Conference on Work-Related Musculoskeletal Disorders involving 26 sonography related professional organizations, accredited bodies, and manufacturers. This collaborative effort updated the 2003 standards and produced a detailed resource to assist in the reduction of WRMSDs among users of sonographic equipment. These overseeing bodies, organizations and sonographers are all in agreement that attrition rate secondary to musculoskeletal discomforts is much too prevalent in our industry. In this seminal report they give responsibilities and details of action to manufacturers, employers, and sonographers. Leading manufacturers of the more detailed and professional quality cardiac sonography systems require using small grocery cart sized consoles equipped with an upper monitor, mid-level keyboard and lower-level computer system. This console is pushed about on its 4 wheels to various locations within the hospital where a probe is pressed against the patient's chest wall and images transferred through a cord connecting the probe to the computer. It is recommended that the system console not require more than 50 pounds of push pull force with the ability to angulate the monitoring screen as well as the keyboard positioning in a manner to maintain neutral posture during the exam. Further recommendations include light weight transducers and cables, adjustable chair, and exam table with height adjustable to maintain arm abduction of less than 30° as well as a drop away or cut out section on the edge of the table to allow less stressful wrist and hand positioning when imaging the apical region of the chest wall. On behalf of the employer the committee suggests that a culture of safety be a shared commitment between management and employees to ensure safety in the work environment. Effective Jan. 1, 2017, revised occupational safety and health administration (OSHA) requirements for reporting include provisions that encourage workers to report work-related injuries or illnesses to their employers and prohibit employers from retaliating against workers for making those reports. Employers are encouraged to maintain ideal workplace conditions, scheduling breaks and anonymous input of symptoms from their sonographers. Sonographers should follow and be aware of best practices to reduce risk of developing WRMSDs. Sonographers are encouraged to be aware of and proactive in managing personal risks by being mindful of job specific risk, signs, and symptoms for WRMSD. Sonographer should employ muscle recovery time throughout the day.

(12) Point-Of-Care ultrasound systems are smaller, portable, compact ultrasound devices used to image the patient at the site of initial care such as in the emergency room, by paramedic or anesthesiologist. These systems are light weight hand-held and configured to be used without a cart or console but rather images are transferred to an iPad or smart phone. These systems by and large are not used for more detailed studies of professionally trained sonographers but rather give some degree of insight and understanding at the moment of need to immediately help guide medical planning. Since such devices are used by non-traditionally trained sonographers, they have similar risk factors and potential to increase exposure to awkward postures and increased observation of WRMSDs.

(13) To date, there has never been an accessory which could be seamlessly implemented into the regular workflow and profoundly reduce the chronic musculoskeletal trauma endured by full-time sonographers.

(14) Accordingly, there is a need for an ultrasonic transducer probe holder suitable for holding an ultrasonic transducer probe of a medical ultrasound system in place against the skin of a patient as an ultrasound technician manually releases the probe.

SUMMARY OF THE INVENTION

(15) The present invention is directed to an ultrasonic transducer probe holder suitable for holding an ultrasonic transducer probe of a medical ultrasound system in place against the skin of a patient as an ultrasound technician manually releases the probe. The ultrasonic transducer probe holder may include a grip portion and a skirt portion. The grip portion may be configured for deployment on the ultrasonic transducer probe. The grip portion may provide a comfortable and secure grip for the ultrasound technician as the technician applies the ultrasonic transducer probe against the skin of the patient. The ultrasonic transducer probe with the grip portion thereon may be insertable in the skirt portion of the ultrasonic transducer probe holder. The skirt portion may be configured to hold or support the ultrasonic transducer probe in an upward-standing position while forming a vacuum seal against the skin of the patient. A stand assembly may be configured for mounting on a bed, table, or other support on which the patient reclines. The stand assembly may be configured to hold or support the ultrasonic transducer probe holder as the holder remains deployed on the ultrasonic transducer probe. Accordingly, the stand assembly may maintain the ultrasonic transducer probe in place against the patient's skin to reduce or eliminate the manual pressure required for the technician to maintain the ultrasonic transducer probe in place during an echocardiography or other ultrasound procedure. The ultrasonic transducer probe holder may prevent carpal tunnel syndrome and other repetitive stress injuries in the hand of the technician.

(16) In an illustrative implementation of the invention, an ultrasonic transducer probe holder suitable for holding an ultrasonic transducer probe of a medical ultrasound system in place against the skin of a patient as an ultrasound technician manually releases the probe may include a grip portion and a skirt portion. The ultrasonic transducer probe of the medical ultrasound system may include a probe handle. A transducer probe may extend from the probe handle. The grip portion of the ultrasonic transducer probe holder may be configured for deployment around the probe handle of the ultrasonic transducer probe. The grip portion may provide a comfortable and secure grip for the ultrasound technician as the technician applies the ultrasonic transducer probe against the skin of the patient. The ultrasonic transducer probe with the grip portion thereon may be insertable in the skirt portion of the ultrasonic transducer probe holder. The skirt portion may contain, enclose or surround the transducer probe of the ultrasonic transducer probe. A stand assembly may be configured for mounting on a bed, table, or other support on which the patient reclines. The stand assembly may be configured to hold or support the ultrasonic transducer probe holder as the holder remains deployed on the ultrasonic transducer probe. Accordingly, the stand assembly may maintain the ultrasonic transducer probe in place against the patient's skin to reduce or eliminate the manual pressure required for the technician to maintain the ultrasonic transducer probe in place during an echocardiography or other ultrasound procedure. The ultrasonic transducer probe holder may prevent carpal tunnel syndrome and other repetitive stress injuries in the hand of the technician.

(17) In a second aspect, the grip portion of the ultrasonic transducer probe holder may include a grip portion wall. A grip portion interior may be formed by the grip portion wall. The grip portion interior may be suitably sized and configured to contain the probe handle of the ultrasonic transducer probe.

(18) In another aspect, the grip portion wall of the grip portion may include at least one elastic or stretchable material. The grip portion wall may be configured to substantially conform to the diameter or width of the probe handle of the ultrasonic transducer probe.

(19) In another aspect, the grip portion may include a pair of mating or interfacing grip portion sections. The grip portion sections may be configured to receive and contain opposite sides of the probe handle of the ultrasonic transducer probe.

- (20) In another aspect, the at least one elastic or stretchable material of the grip portion wall may include silicone.
- (21) In another aspect, the ultrasonic transducer probe may include a probe shaft which connects the transducer probe to the probe handle. The grip portion wall of the grip portion may include a grip portion base which is configured to engage the probe shaft and a grip portion middle section extending from the grip portion base and a grip portion apex extending from the grip portion middle section. The grip portion middle section and the grip portion apex of the grip portion may be configured to engage the probe handle of the ultrasonic transducer probe.
- (22) In another aspect, the probe handle may have a greater diameter or width than the probe shaft of the ultrasonic transducer probe. The grip portion middle section of the grip portion wall may be expandable to a greater diameter or width than that of the grip portion base and the grip portion apex of the grip portion wall to accommodate the relatively greater diameter or width of the probe handle.
- (23) In another aspect, the grip portion base of the grip portion wall may have a grip portion base edge. In placement of the ultrasonic transducer probe in the skirt portion, the grip portion base edge of the grip portion wall may be configured to engage the skirt portion.
- (24) In another aspect, a grip portion base opening may be circumscribed by the grip portion base edge of the grip portion base.
- (25) In another aspect, the grip portion apex of the grip portion wall may have a grip portion apex edge which is opposite the grip portion base edge of the grip portion base.
- (26) In another aspect, a grip portion apex opening may be circumscribed by the grip portion apex edge of the grip portion apex.
- (27) In another aspect, the skirt portion may be generally funnel-shaped.
- (28) In another aspect, the skirt portion of the ultrasonic transducer probe holder may include a skirt portion wall. A skirt portion interior may be formed by the skirt portion wall. The skirt portion interior may be suitably sized and configured to contain the transducer probe of the ultrasonic transducer probe.
- (29) In another aspect, the skirt portion wall of the skirt portion may include at least one elastic or stretchable material. The skirt portion wall may flare or extend outwardly from the transducer probe of the ultrasonic transducer probe.
- (30) In another aspect, the at least one elastic or stretchable material of the skirt portion wall may include silicone.
- (31) In another aspect, the skirt portion wall of the skirt portion may include a relatively wide skirt portion base and a relatively narrow skirt portion apex extending from the skirt portion base.
- (32) In another aspect, the skirt portion base of the skirt portion may have a skirt portion base opening through which the transducer probe is exposed to facilitate contact of the transducer probe with the skin of the patient.
- (33) In another aspect, the skirt portion base of the skirt portion wall may include a skin engaging edge which circumscribes the skirt portion base opening of the skirt portion base. The skin engaging edge may be configured to engage the skin of the patient.
- (34) In another aspect, the skirt portion apex of the of the skirt portion wall may have a skirt portion apex opening. The skirt portion apex opening may be configured to interface and communicate with the grip portion base opening of the grip portion base of the grip portion wall.
- (35) In another aspect, the skirt portion apex of the skirt portion wall may have a skirt portion apex edge which circumscribes the skirt portion apex opening. The skirt portion apex edge of the skirt portion apex may be configured to engage or may be continuous with the grip portion base edge on the grip portion base of the grip portion wall.
- (36) In another aspect, the skirt portion wall of the skirt portion may have a skirt portion middle section extending between the skirt portion base and the skirt portion apex. The skirt portion middle section may gradually expand in width or diameter from the skirt portion apex to the skirt

portion base.

(37) In another aspect, the stand assembly may include a stand base. A cradle base may be supported by the stand base. An elongated cradle gooseneck may extend from the cradle base. A cradle may be supported by the cradle gooseneck. The ultrasonic transducer probe with the probe holder thereon may be supported by the cradle during use of the probe.

(38) In another aspect, the stand assembly may include a lower base member. At least one base arm may extend from the lower base member. An tipper base member may be supported by the base arm. The cradle base may be supported by the upper base member.

(39) In another aspect, the cradle base may be adjustably mounted with respect to the upper base member.

(40) In another aspect, an elongated base slot may extend through the upper base member of the stand base. The cradle base may adjustably engage the base slot.

(41) In another aspect, the upper base member of the stand base may include at least one electrical outlet.

(42) These and other objects, features, and advantages of the present invention will become more readily apparent from the attached drawings and the detailed description of the preferred embodiments, which follow.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The preferred embodiments of the invention will hereinafter be described in conjunction with the appended drawings provided to illustrate and not to limit the invention, where like designations denote like elements, and in which:

(2) FIG. 1 presents a perspective view of a typical grip portion of an ultrasonic transducer probe holder in accordance with an illustrative embodiment of the present invention, with the grip portion deployed in place on an ultrasonic transducer probe in typical application of the ultrasonic transducer probe holder;

(3) FIG. 2 presents an exploded perspective view of the illustrative ultrasonic transducer probe holder as the grip portion is being deployed in place on the ultrasonic transducer probe in typical application of the ultrasonic transducer probe holder;

(4) FIG. 3 presents an exploded perspective view of the illustrative ultrasonic transducer probe holder, with the grip portion deployed on the ultrasonic transducer probe and further illustrating typical placement of the ultrasonic transducer probe in the skirt portion of the ultrasonic transducer probe holder;

(5) FIG. 4 presents a perspective view illustrating typical sliding deployment of the skirt portion on the ultrasonic transducer probe;

(6) FIG. 5 presents a perspective view of the grip portion deployed on the ultrasonic transducer probe and the ultrasonic transducer probe placed in the skirt portion of the ultrasonic transducer probe holder and further illustrating typical downward application of the skirt portion against the skin of a patient in typical application of the ultrasonic transducer probe holder;

(7) FIG. 6 presents a bottom perspective view of the grip portion of the ultrasonic transducer probe holder deployed on the ultrasonic transducer probe, more particularly illustrating a vacuum tube opening and a gel tube opening in the grip portion and a vacuum tube and a gel tube extending through the respective vacuum tube opening and gel tube opening in typical application of the ultrasonic transducer probe holder;

(8) FIG. 7 presents an exploded top perspective view of a typical stand assembly configured to hold or support the probe holder as the probe holder remains deployed on the ultrasonic transducer probe;

(9) FIG. 8 presents an exploded bottom perspective view of the stand assembly;

(10) FIG. 9 presents an exploded side perspective view of the stand assembly, more particularly illustrating a typical technique for attaching the cradle base to the stand base of the stand assembly;

(11) FIG. 10 presents a side view of a typical stand base of the stand assembly, more particularly vertical and horizontal adjustment capability of the stand base;

(12) FIG. 11 presents a top perspective view of the stand assembly with the ultrasonic transducer probe holder deployed on the ultrasonic transducer probe and the ultrasonic transducer probe placed in a cradle of the stand assembly;

(13) FIG. 12 presents a side view of the stand assembly with ultrasonic transducer probe and probe holder illustrated in FIG. 11, with the stand base of the stand assembly deployed on a table, bed or other support in typical application of the ultrasonic transducer probe holder;

(14) FIG. 13 presents a side view (top image) and a top view (bottom image) of the upper base member of the stand base, more particularly illustrating a battery compartment in the upper base member, a battery in the battery compartment and an electrical outlet electrically interfacing with the battery compartment;

(15) FIG. 14 presents a longitudinal sectional view of the ultrasonic transducer probe with the grip portion deployed in place on the probe, the probe deployed in place in the skirt portion of the ultrasonic transducer probe holder and the skirt portion engaging the skin of a patient in typical application of the ultrasonic transducer probe holder;

(16) FIG. 15 presents an enlarged sectional view of the skirt portion engaging the skin of the patient and the ultrasonic transducer probe in the skirt portion, more particularly illustrating selective discharge of conductive gel through a gel tube and from a gel tube end in the probe onto the skin of the patient;

(17) FIG. 16 presents a side view of an ultrasonic transducer probe holder and a typical stand assembly for the probe holder in accordance with an alternative illustrative embodiment of the present invention;

(18) FIG. 17 presents a front view of the illustrative ultrasonic transducer probe holder and stand assembly illustrated in FIG. 16;

(19) FIG. 18 presents a rear perspective view of the ultrasonic transducer probe holder illustrated in FIG. 16, with the stand assembly deployed in a raised position;

(20) FIG. 19 presents a front perspective view of the ultrasonic transducer probe holder illustrated in FIG. 16, with the stand assembly of the probe holder deployed in a lowered position;

(21) FIG. 20 presents a side view of the ultrasonic transducer probe holder illustrated in FIG. 16, detached from the stand assembly (not illustrated);

(22) FIG. 21 presents a front view of the ultrasonic transducer probe holder illustrated in FIG. 20;

(23) FIG. 22 presents a side view of the ultrasonic transducer probe holder, with a mount clevis (not illustrated) for the stand assembly detached from the probe holder;

(24) FIG. 23 presents a top perspective view of the ultrasonic transducer probe holder illustrated in FIG. 22;

(25) FIG. 24 presents a longitudinal sectional view, taken along section lines A-A in FIG. 25, of the ultrasonic transducer probe holder assembled on an ultrasonic transducer probe in typical application of the probe holder;

(26) FIG. 25 presents a front view of the ultrasonic transducer probe holder illustrated in FIG. 24;

(27) FIG. 26 presents a bottom perspective view of the ultrasonic transducer probe holder and ultrasonic transducer probe;

(28) FIG. 27 presents a side view of the partially assembled ultrasonic transducer probe holder, with a grip portion section of the grip portion assembled in place on the ultrasonic transducer probe, more particularly illustrating a typical ball socket ring technique suitable for attaching the probe holder mount clevis to the ball socket of the probe holder;

(29) FIG. 28 presents a front view of the partially assembled ultrasonic transducer probe holder

illustrated in FIG. 27;

(30) FIG. 29 presents a longitudinal sectional view, taken along section lines 29-29 in FIG. 28, of the ultrasonic transducer probe holder, assembled on the ultrasonic transducer probe, and the mount clevis attached to the ball socket of the probe holder;

(31) FIG. 30 presents an enlarged longitudinal sectional view of the grip portion of the ultrasonic transducer probe holder assembled on the ultrasonic transducer probe illustrated in FIG. 29;

(32) FIG. 31 presents a longitudinal sectional view of the assembled ultrasonic transducer probe holder and ultrasonic transducer probe with the probe applied to the skin of a patient in typical application of the probe holder;

(33) FIG. 32 presents an exploded perspective view of the ultrasonic transducer probe and the disassembled ultrasonic transducer probe holder;

(34) FIG. 33 presents a top perspective view of the assembled ultrasonic transducer probe holder, with the mount clevis attached to the ball socket of the probe holder;

(35) FIG. 34 presents a bottom perspective view of the assembled ultrasonic transducer probe holder and ultrasonic transducer probe;

(36) FIG. 35 presents a block diagram of a typical gel distribution circuit suitable for distributing gel from a pump and supply mechanism to the skirt portion of the ultrasonic transducer probe holder;

(37) FIG. 36 presents a block diagram of the gel distribution circuit with a pressure transducer between a check valve and the skirt portion of the ultrasonic transducer probe holder;

(38) FIG. 37 presents a block diagram of a typical vacuum pressure control, gel quantity control and gel temperature control for the pump and supply mechanism of the ultrasonic transducer probe holder; and

(39) FIGS. 38-52 present perspective views of the ultrasonic transducer probe holder in typical application of the probe holder.

(40) Like reference numerals refer to like parts throughout the several views of the drawings.

Detailed Description

(41) The following detailed description is merely exemplary in nature and is not intended to limit the described embodiments or the application and uses of the described embodiments. As used herein, the word “exemplary” or “illustrative” means “serving as an example, instance, or illustration.” Any implementation described herein as “exemplary” or “illustrative” is not necessarily to be construed as preferred or advantageous over other implementations. All of the implementations described below are exemplary implementations provided to enable persons skilled in the art to make or use the embodiments of the disclosure and are not intended to limit the scope of the disclosure, which is defined by the claims. For purposes of description herein, the terms “upper”, “lower”, “left”, “rear”, “right”, “front”, “vertical”, “horizontal”, and derivatives thereof shall relate to the invention as oriented in FIG. 1. Furthermore, there is no intention to be bound by any expressed or implied theory presented in the preceding technical field, background, brief summary or the following detailed description. It is also to be understood that the specific devices and processes illustrated in the attached drawings, and described in the following specification, are simply exemplary embodiments of the inventive concepts defined in the appended claims. Hence, specific dimensions and other physical characteristics relating to the embodiments disclosed herein are not to be considered as limiting, unless the claims expressly state otherwise.

(42) Shown throughout the figures, the present invention is directed toward an ultrasonic transducer probe holder suitable for holding an ultrasonic transducer probe of a medical ultrasound system in place against the skin of a patient as an ultrasound technician manually releases the probe.

(43) Referring initially to FIGS. 1-6, 11, 12, 14 and 15, an ultrasonic transducer probe holder, hereinafter probe holder **100**, is illustrated in accordance with an exemplary embodiment of the present invention. As will be hereinafter further described, the probe holder **100** may be configured

for deployment on an ultrasonic transducer probe **150** of a medical ultrasound system (not illustrated). The medical ultrasound system may be standard or conventional, typically having a computer with a user interface and a display. A probe cable **158** may connect the ultrasonic transducer probe **150** to the computer. Accordingly, in typical operation of the medical ultrasound system, which will be hereinafter further described, the ultrasonic transducer probe **150** may be placed against the skin **164** of a patient **162** typically as the patient **162** reclines on a table or bed (not illustrated). As illustrated in FIGS. **11** and **12**, a stand assembly **200** may be configured for mounting on a bed, table, or other support **248** (FIG. **12**) on which the patient reclines. The stand assembly **200** may be configured to hold or support the probe holder **100** as the probe holder **100** remains deployed on the ultrasonic transducer probe **150**. The ultrasonic transducer probe **150** may generate sonic waves which impinge against and through the skin **164** of the patient **162** and receive the reflected sonic waves which echo from the organ or tissue being imaged. The computer may generate 2D or 3D images of the imaged organ or tissue of the patient on the display, typically for diagnostic purposes.

(44) As shown for instance in FIG. **3**, the probe holder **100** may include a grip portion **102** and a skirt portion **126**. The grip portion **102** may be configured for deployment on the ultrasonic transducer probe **150**. The grip portion **102** may provide a comfortable and secure grip for an ultrasound technician (not illustrated) as the technician applies the ultrasonic transducer probe **150** against the skin **164** of the patient **162**, as illustrated in FIGS. **14** and **15**. The ultrasonic transducer probe **150** with the grip portion **102** thereon may be insertable in the skirt portion **126** of the probe holder **100**, as illustrated in FIG. **3**. The skirt portion **126** may be configured to hold or support the ultrasonic transducer probe **150** in an upward-standing position while forming a vacuum seal against the skin **164** of the patient **162**. Accordingly, the skirt portion **126** may maintain the ultrasonic transducer probe **150** in place against the patient's skin **164** to reduce the manual pressure required for the technician to maintain the ultrasonic transducer probe **150** in place during an echocardiography or other procedure.

(45) As illustrated in FIG. **2**, in some applications, the ultrasonic transducer probe **150** of the medical ultrasound system may include a probe handle **152**. A probe base **156** may extend from the probe handle **152**. A colored, typically green light **168** may be provided on the ultrasonic transducer probe **150** such as to indicate the operational orientation of the probe **150**. The grip portion **102** of the probe holder **100** may be configured for deployment around the probe handle **152** of the ultrasonic transducer probe **150**. When the ultrasonic transducer probe **150** with the grip portion **102** thereon is inserted in the skirt portion **126**, the skirt portion **126** may contain, enclose, or surround the probe base **156** of the ultrasonic transducer probe **150**.

(46) In some embodiments, the grip portion **102** of the probe holder **100** may include a grip portion wall **104**. As illustrated in FIG. **14**, a grip portion interior **114** may be formed by the grip portion wall **104**. The grip portion interior **114** may be suitably sized and configured to contain the probe handle **152** of the ultrasonic transducer probe **150**.

(47) In some embodiments, the grip portion wall **104** of the grip portion **102** may include at least one elastic or stretchable material. Accordingly, the grip portion wall **104** may be configured to substantially conform to the diameter or width of the probe handle **152** of the ultrasonic transducer probe **150**. As illustrated in FIG. **2**, in some embodiments, the grip portion **102** may include a pair of mating or interfacing grip portion sections **112**. The grip portion sections **112** may be configured to receive and contain opposite sides of the probe handle **152** of the ultrasonic transducer probe **150**. In some embodiments, the grip portions **112** may have a hinged clamshell design, or may be configured to separately and detachably engage each other according to the knowledge of those skilled in the art.

(48) In some embodiments, the at least one elastic or stretchable material of the grip portion wall **104** may include silicone, for example and without limitation. In other embodiments, the material may include stretchable plastic and/or rubber, for example and without limitation. In various

applications, the ultrasonic transducer probe **150** may have different sizes, shapes and configurations. Accordingly, the typically elastic construction of the grip portion wall **104** may facilitate deployment of the grip portion **102** on ultrasonic transducer probes **150** having the different sizes, shapes and configurations. In some embodiments, the grip portion wall **104** of the grip portion **102** may be fabricated to match the specific size and configuration of the ultrasonic transducer probe **150**.

(49) As further illustrated in FIG. 2, in some applications of the probe holder **100**, the ultrasonic transducer probe **150** may include a probe shaft **154** which connects the probe base **156** to the probe handle **152**. The grip portion wall **104** of the grip portion **102** may include a grip portion base **106** which is configured to engage the probe shaft **154**. A grip portion middle section **110** may extend from the grip portion base **106**. A grip portion apex **108** may extend from the grip portion middle section **110**. The grip portion middle section **110** and the grip portion apex **108** of the grip portion **102** may be configured to engage the probe handle **152** of the ultrasonic transducer probe **150**.

(50) In some applications, the probe handle **152** of the ultrasonic transducer probe **150** may have a greater diameter or width than that of the probe shaft **154** of the ultrasonic transducer probe **150**. Thus, the grip portion middle section **110** of the grip portion wall **104** may be expandable to a greater diameter or width than that of the grip portion base **106** and the grip portion apex **108** of the grip portion wall **104** to accommodate the relatively greater diameter or width of the probe handle **152**.

(51) As illustrated in FIG. 14, a vacuum tube **180** and a gel tube **182** may be insertable into the ultrasonic transducer probe **150**. The vacuum tube **180** may be connected to a vacuum system (not illustrated). The gel tube **182** may be connected to a pump and supply mechanism (not illustrated) which contains a conductive gel **160** (FIG. 15). The vacuum tube **180** may terminate in a vacuum tube end **184** in the ultrasonic transducer probe **150**. As illustrated in FIG. 15, an open gel tube end **186** may terminate the gel tube **182** at the lower end of the probe base **156**. Accordingly, vacuum pressure may be selectively applied to the ultrasonic transducer probe **150** through the vacuum tube **180**. A selected quantity or volume of the conductive gel **160** may be dispensed from the probe base **156** of the ultrasonic transducer probe **150** through the gel tube **182** and the gel tube end **186**, respectively.

(52) In some embodiments, the grip portion base **106** of the grip portion wall **104** may have a grip portion base edge **116**. A grip portion base opening **120** may be circumscribed by the grip portion base edge **116** of the grip portion base **106**. The grip portion base opening **120** may communicate with the grip portion interior **114** of the grip portion **102**. In placement of the ultrasonic transducer probe **150** in the skirt portion **126**, as illustrated in FIG. 3, the grip portion base edge **116** of the grip portion wall **104** may be configured to engage the skirt portion **126**.

(53) The grip portion apex **108** of the grip portion wall **104** may have a grip portion apex edge **118** which is opposite the grip portion base edge **116** of the grip portion base **106**. A grip portion apex opening **122** may be circumscribed by the grip portion apex edge **118** of the grip portion apex **108**. The grip portion apex opening **122** may be sized and configured to facilitate passage of the probe cable **158** from the ultrasonic transducer probe **150** to the computer **172** of the medical ultrasound system **170**.

(54) As illustrated in FIGS. 2 and 14, a vacuum tube opening **190** may extend through the grip portion wall **104** at the grip portion apex **108** of the grip portion **102**. The vacuum tube opening **190** may accommodate the vacuum tube **180** as the vacuum tube **180** extends from the vacuum system (not illustrated) to the ultrasonic transducer **150**. A gel tube opening **192** may extend through the grip portion wall **104** at the grip portion apex **108**. The gel tube opening **192** may accommodate the gel tube **182** as the gel tube **182** extends from the pump and supply mechanism (not illustrated) which contains the conductive gel **160** to the ultrasonic transducer **150**.

(55) The skirt portion **126** of the probe holder **100** may be generally funnel-shaped. In some

embodiments, the skirt portion **126** may include a skirt portion wall **128**. A skirt portion interior **142**, where the vacuum is applied, may be formed by the skirt portion wall **128**. The skirt portion interior **142** may be suitably sized and configured to contain the probe base **156** of the ultrasonic transducer probe **150**.

(56) In some embodiments, the skirt portion wall **128** of the skirt portion **126** may include at least one elastic or stretchable material. For example and without limitation, in some embodiments, the at least one elastic or stretchable material of the skirt portion wall **128** may include silicone. In other embodiments, the material may include stretchable plastic and/or rubber, for example and without limitation. As illustrated in FIG. 5, when the ultrasonic transducer probe **150** is deployed in place in the skirt portion **126**, the skirt portion wall **128** may flare or extend outwardly from the probe base **156** of the ultrasonic transducer probe **150**.

(57) In some embodiments, the skirt portion wall **128** of the skirt portion **126** may include a relatively wide skirt portion base **130**. A relatively narrow skirt portion apex **132** may extend from the skirt portion base **130**. As illustrated in FIG. 14, the skirt portion base **130** of the skirt portion **126** may have a skirt portion base opening **144** through which the probe base **156** may be exposed to facilitate contact of the probe base **156** with the skin **164** of the patient **162**.

(58) As further illustrated in FIG. 5, the skirt portion base **130** of the skirt portion wall **128** may include a skin engaging edge **138**. The skin engaging edge **138** may circumscribe the skirt portion base opening **144** of the skirt portion base **130**. The skin engaging edge **138** may be configured to engage the skin **164** of the patient **162** in typical application of the probe holder **100**.

(59) The skirt portion apex **132** of the skirt portion wall **128** may have a skirt portion apex opening **146**. The skirt portion apex opening **146** may be configured to interface and communicate with the grip portion base opening **120** of the grip portion base **106** of the grip portion wall **104**.

(60) The skirt portion apex **132** of the skirt portion wall **128** may have a skirt portion apex edge **140** which circumscribes the skirt portion apex opening **146**. The skirt portion apex edge **140** of the skirt portion apex **132** may be configured to engage or face the grip portion base edge **116** on the grip portion base **106** of the grip portion wall **104** in deployment of the skirt portion **126** on the ultrasonic transducer probe **150**. Alternatively, in some embodiments, the skirt portion wall **128** of the skirt portion **126** may be fabricated in one piece or may be continuous with the grip portion wall **104** of the grip portion **102** according to the knowledge of those skilled in the art.

(61) In some embodiments, the skirt portion wall **128** of the skirt portion **126** may have a skirt portion middle section **136** which extends between the skirt portion base **130** and the skirt portion apex **132**. The skirt portion middle section **136** may gradually expand in width or diameter from the skirt portion apex **132** to the skirt portion base **130**.

(62) As illustrated in FIGS. 7-13 of the drawings, in some embodiments, a stand assembly **200** may be configured to hold or support the probe holder **100** as the probe holder **100** remains deployed on the ultrasonic transducer probe **150**. The stand assembly **200** may support the ultrasonic transducer probe **150** as the ultrasonic transducer probe **150** is applied to the skin **164** of the patient **162**. The stand assembly **200** may include a stand base **202**. A cradle base **220** may be supported by the stand base **202**. An elongated cradle gooseneck **224** may extend from the cradle base **220**. A cradle **226** may be supported by the cradle gooseneck **224**. As illustrated in FIG. 11, the cradle **226** may be suitably sized and configured to support the ultrasonic transducer probe **150** with the probe holder **100** in various positions on the patient as selected by the sonographer.

(63) As illustrated in FIG. 12, the stand base **202** may be configured to engage a support **248** for support thereby throughout use of the ultrasonic transducer probe **150**. For example and without limitation, in some embodiments, the support **248** may include a frame or other structural component on a bed or table on which the patient **162** reclines. In some embodiments, the stand base **202** may have a lower base member **204**. At least one base arm **208** may extend from the lower base member **204**. An upper base member **206** may be supported by the base arm **208**. The cradle base **220** may be supported by the upper base member **206** of the stand base **202**. The stand

base **202** may be configured to receive the support **248** between the lower base member **204** and the upper base member **206**.

(64) As illustrated in FIG. **10**, in some embodiments, the position of the upper base member **206** may be selectively adjustable with respect to the lower base member **204** of the stand base **202** according to the knowledge of those skilled in the art. For example and without limitation, in some embodiments, the base arms **208** may be telescopically adjustable for the purpose. Accordingly, as illustrated in FIG. **12**, the stand base **202** may be configured to accommodate supports **248** having different thicknesses between the lower base member **204** and the upper base member **206** by adjustment of the upper base member **206** with respect to the lower base member **204**.

(65) In some embodiments, the cradle base **220** may be adjustably mounted with respect to the upper base member **206** of the stand base **202**. Accordingly, as illustrated in FIG. **7**, in some embodiments, an elongated base slot **212** may extend through the upper base member **206** of the stand base **202**. The cradle base **220** may adjustably engage the base slot **212**. As illustrated in FIGS. **8** and **9**, in some embodiments, a threaded cradle nipple **232** may extend downwardly from the cradle base **220**. The cradle nipple **232** may extend downwardly through the base slot **212**. As illustrated in FIG. **9**, a cradle base plate **230** may have a central plate opening **234** which receives the cradle nipple **232**. The cradle base plate **230** may be deployed on the cradle nipple **232** beneath the upper base member **206**. In some embodiments, a cradle disk **236**, having a central disk opening **238** and fabricated of rubber or other resilient material, may be placed on the cradle nipple **232** between the cradle base plate **230** and the cradle base **220**. A cradle cap **242** may be threaded and tightened on the extending portion of the cradle nipple **232** to secure the cradle base **220** against the upper surface of the upper base member **206**. Accordingly, the cradle cap **242** may be selectively loosened on the cradle nipple **232** to facilitate movement of the cradle base **220** with respect to the upper base member **206** as the cradle nipple **232** traverses the base slot **212**. The cradle cap **242** may be subsequently tightened on the cradle nipple **232** to immobilize the cradle base **220** at the selected position on the upper base member **206**.

(66) In some embodiments, a cradle base lock lever **222** may be provided on the cradle base **220**. The cradle base lock lever **222** may be selectively deployable in an unlock position (FIG. **9**) and a lock position (FIGS. **11** and **12**). The cradle base lock lever **222** may operably engage the cradle nipple **232** to facilitate selective advancement (in the unlock position) and retraction (in the lock position) of the cradle nipple **232** with respect to the cradle base **220** responsive to positional manipulation of the cradle base lock lever **222**, according to the knowledge of those skilled in the art. Accordingly, movement of the cradle base **220** with respect to the upper base member **206** may be accomplished by manipulation of the cradle base lock lever **222** to the unlock position. Securement of the cradle base **220** at the selected position on the upper base member **206** may be accomplished by manipulation of the cradle base lock lever **222** from the unlock position to the lock position.

(67) As illustrated in FIG. **13**, in some embodiments, the upper base member **206** of the stand base **202** may include at least one electrical outlet **216**. The electrical outlet **216** may be provided along an edge, as illustrated, or in any other accessible position of the upper base member **206**. A battery compartment **218** may be provided in the upper base member **206**. The battery compartment **218** may electrically interface with the electrical outlet **216**. The battery compartment **218** may be sized and configured to contain at least one battery **214** which supplies electrical current to the electrical outlet **216**. Accordingly, the electrical outlet **216** may provide electrical current to the ultrasonic transducer probe **150** and/or to one or more electrical accessories during use of the ultrasonic transducer probe **150**.

(68) As further illustrated in FIG. **13**, in some embodiments, at least one gel container storage compartment **250** may be provided in the upper base member **206**, as illustrated, and/or in the lower base member **204** of the stand base **202**. The gel container storage compartment **250** may be suitably sized and configured to contain at least one gel container **1** which contains a supply of the

conductive gel **160**.

(69) As illustrated in FIGS. **1-6**, **14** and **15**, in typical application of the probe holder **100**, the grip portion **102** may initially be deployed in place on the probe handle **152** and the probe shaft **154** of the ultrasonic transducer probe **150**. Accordingly, the grip portion inner wall **104** may be stretched over the ultrasonic transducer probe **150** as the probe base **156** and then the probe shaft **154** and the probe handle **152**, respectively, are inserted through the grip portion base opening **120** in the grip portion base **106**. The grip portion wall **104** may then be released to recoil snugly against the probe handle **152** and the probe shaft **154**. As illustrated in FIG. **2**, in some applications, the grip portion **102** may be deployed by placing the grip portion sections **112** around the respective sides of the ultrasonic transducer probe **150** and attaching the grip portions **112** to each other. As illustrated in FIGS. **6**, the vacuum tube **180** and the gel tube **182** may be inserted through the respective vacuum tube opening **190** and gel tube opening **192** in the grip portion apex **108** of the grip portion **102**. The vacuum tube **180** may be connected to the vacuum system (not illustrated). The gel tube **182** may be connected to the pump and supply mechanism (not illustrated) which contains a supply of the conductive gel **160** (FIG. **15**). The probe cable **158** which connects the ultrasonic transducer probe **150** to the computer of the medical ultrasound system may extend through the grip portion apex opening **122** at the grip portion apex edge **118** of the grip portion apex **108**.

(70) As illustrated in FIG. **4**, the skirt portion **126** may be deployed in place on the ultrasonic transducer probe **150** beneath or adjacent to the grip portion **102**. Accordingly, the skirt portion **126** may be stretched and initially slid over and then released against the probe base **156** of the ultrasonic transducer probe **150**. As illustrated in FIGS. **11** and **12**, the ultrasonic transducer probe **150** with the probe holder **100** thereon may be placed in the cradle **226** of the stand assembly **200**. The position of the cradle **226** may be selected by adjustment of the lower base member **204** and tipper base member **206** of stand base **202** with respect to each other (FIG. **10**), horizontal adjustment of the cradle base **202** with respect to the upper base member **206** (FIG. **10**) and universal adjustment of the cradle gooseneck **224**.

(71) The patient **162** may recline on the bed, table, or other support **248** (FIG. **12**) on which the stand assembly **200** is deployed. The ultrasound technician may grasp the grip portion **102** of the probe holder **100** as the probe holder **100** remains in place in the cradle **226** of the stand assembly **200**. As the ultrasound technician locates a position on the skin **164** of the patient **162** and releases the probe holder **100**, the cradle gooseneck **224** may enable the probe base **156** to remain in position. As illustrated in FIGS. **14** and **15**, the skin engaging edge **138** on the skirt portion base **130** of the skirt portion **126** may next be placed against the skin **164** of the patient **162** at the selected position on the skin **164**. For example and without limitation, in an echocardiography procedure, the skirt portion **126** may be placed on the torso of the patient **162** as the patient **162** lies in a supine position on the support **248**. As illustrated by the arrow in FIG. **5**, the technician may initially manually apply downward force on the grip portion **102** such that an airtight suction seal is formed between the skin engaging edge **138** of the skirt portion base **130** and the surface of the patient's skin **164**. The technician may then release the grip portion **102** as cradle **226** on the cradle gooseneck **224** continues to support the probe holder **100** and the ultrasonic transducer probe **150** held therein typically in an upward-standing position on the skin **164** of the patient **162**.

(72) As the ultrasonic transducer probe **150** remains suspended by the cradle **226** of the stand assembly **200** without touch by the ultrasound technician, removal of the hand of the technician from the probe holder **100** typically activates the vacuum system connected to the ultrasonic transducer probe **150** through the vacuum tube **180**. The vacuum system may generate closer and tighter approximation of the probe base **156** to the skin **164** of the patient **162**, thereby simulating direct pressure of the probe base **156** into the chest of the patient **162** that would otherwise be applied by the hand of the technician. This hands-free mechanism of operating the ultrasonic transducer probe **150** will greatly reduce the musculoskeletal trauma so prevalent among ultrasound technicians today.

(73) As illustrated in FIG. 15, the conductive gel 160 may be dispensed through the gel tube 182 from the open gel tube end 186 at the lower end of the probe base 156 and onto the patient's skin 164. The probe base 156 of the ultrasonic transducer probe 150 may generate sonic waves which impinge against and through the skin 164 of the patient 162 and receive the reflected sonic waves which echo from the organ or tissue being imaged. The computer may generate 2D or 3D images of the imaged organ or tissue of the patient 162 on the display, typically in the conventional manner, for diagnostic purposes. As illustrated in FIGS. 11 and 12, when not in use, the ultrasonic transducer probe 150 with the probe holder 100 thereon may remain in place in the cradle 226 of the stand assembly 200.

(74) It will be appreciated by those skilled in the art that, by enabling the ultrasound technician to manually release the ultrasonic transducer probe 150 during much of the ultrasound procedure, the probe holder 100 and stand assembly 200 may prevent carpal tunnel syndrome and other repetitive stress injuries in the hand of the technician. In some embodiments, the grip portion 102 may be manually rotatable with respect to the skirt portion 126 as the grip portion base edge 116 on the grip portion base 106 slides on and with respect to the skirt portion apex edge 140 on the skirt portion apex 132 of the skirt portion 126. This expedient may facilitate corresponding rotation of the probe base 156 with respect to the skin 164 of the patient 162, enabling the technician to obtain different images of the organ or tissues being imaged.

(75) In some applications, after use of the probe holder 100, the ultrasonic transducer probe 150 may be removed from the skirt portion 126, the grip portion 102 removed from the ultrasonic transducer probe 150 and the skirt portion 126 removed from the skin 164 of the patient 162. The grip portion 102 and the skirt portion 126 may be discarded or cleaned and reused.

(76) Referring next to FIGS. 16-52 of the drawings, an alternative illustrative embodiment of the ultrasonic transducer probe holder, hereinafter probe holder 300, is illustrated. Unless otherwise noted, in the probe holder 300, elements which are analogous to the respective elements of the probe holder 100 that was heretofore described with respect to FIGS. 1-15 are designated by the same respective numerals in the 300-399 series in FIGS. 16-52. Accordingly, the same description which was set forth hereinabove with respect to the probe holder 100 in FIGS. 1-15 is incorporated by reference herein in its entirety with respect to the holder 300.

(77) As illustrated in FIGS. 16-19, in some embodiments, a stand assembly 400 may facilitate mounting of the probe holder 300 on the bed, table, or other support 348 (FIGS. 38-45) on which the patient 362 reclines. The stand assembly 400 may facilitate multi-positioning of the probe holder 300 as the ultrasound technician 398 (FIGS. 42-49) operates and manipulates the ultrasonic transducer probe 350 in the probe holder 300 against the skin 364 of the patient 362. The stand assembly 400 may have any design which is suitable for the purpose. For example and without limitation, in some embodiments, the stand assembly 400 may include a stand base 402. The stand base 402 of the stand assembly 400 may be configured to securely engage a structural component of the support 348. Accordingly, the stand base 402 may include a main base member 404. A base member flange 406 may be disposed in spaced-apart relationship to the main base member 404. A base member arm 408 may connect the base member flange 406 to the main base member 404. The stand base 402 may be attached to a rail, panel, or other structural component (not illustrated) on the support 348 by placement of the component between the main base member 404 and the base member flange 406.

(78) In some embodiments, a base adjustment knob 418 may be provided on the main base member 404 of the stand base 402. The base adjustment knob 418 may threadably engage the base member arm 408 according to the knowledge of those skilled in the art to facilitate extension and retraction of the base member arm 408 with respect to the stand base 402 by rotation of the base adjustment knob 418. Accordingly, the spacing between the main base member 404 and the base member arm 408 may be selectively adjusted by rotation of the base adjustment knob 418 to accommodate various thicknesses and facilitate placement and tightening of the stand base 402 on the support

348.

(79) In some embodiments, various controls for the pump and supply mechanism **450** for the gel, the vacuum system **460**, and/or other components of the ultrasonic transducer probe system may be provided on the main base member **404** of the stand base **402**, typically as will be hereinafter described. The controls may interface with the respective components via a wired or wireless connection according to the knowledge of those skilled in the art.

(80) A first stand arm **412** may extend from the stand base **402**. A second stand arm **420** may extend from the first stand arm **412**. The probe holder **300** may be supported by the second stand arm **420**, typically as will be hereinafter described. The stand assembly **400** may be selectively adjustable positionally to facilitate adjustment in the height and orientation of the probe holder **300**. Accordingly, the first stand arm **412** may be selectively adjustable with respect to the stand base **402**. The second stand arm **420** may be selectively adjustable with respect to the first stand arm **412**. The probe holder **300** may be selectively adjustable with respect to the second stand arm **420**.

(81) In some embodiments, the first stand arm **412** of the stand assembly **400** may be elongated with a proximal arm end **414** and a distal arm end **416**. The proximal arm end **414** of the first stand arm **412** may be pivotally attached to the stand base **402**. Accordingly, in some embodiments, a base pivot fastener **410** may pivotally attach the proximal arm end **414** of the first stand arm **412** to the main base member **404** of the stand base **402**. The base pivot fastener **410** may facilitate pivoting of the first stand arm **412** with respect to the stand base **402** typically within a vertical plane.

(82) In some embodiments, the base pivot fastener **410** may be rotatably mounted to the main base member **404** of the stand base **402**. Accordingly, the base pivot fastener **410** may additionally facilitate rotation of the first stand arm **412** along the longitudinal axis of the first stand arm **412**.

(83) The second stand arm **420** of the stand assembly **400** may be elongated with a proximal arm end **422** and a distal arm end **424**. A first pivot fastener **426** may pivotally attach the proximal arm end **422** of the second stand arm **420** to the distal arm end **416** of the first stand arm **412**. The first pivot fastener **426** may facilitate pivoting of the second stand arm **420** with respect to the first stand arm **412** typically within a vertical plane.

(84) A holder mount clevis **432** may extend from the distal arm end **424** of the second stand arm **420**. In some embodiments, a second pivot fastener **428** may pivotally attach the holder mount clevis **432** to the distal arm end **424** of the second stand arm **420**. The holder mount clevis **432** may have any design which is configured to hold or support the probe holder **300**. Accordingly, in some embodiments, the holder mount clevis **432** may have a clevis shaft **434**. A clevis ring **436** may terminate a proximal end of the clevis shaft **434**. The clevis ring **436** may be configured to receive the second pivot fastener **428** for pivotal attachment of the holder mount clevis **432** to the distal arm end **424** of the second stand arm **420**. A ball socket ring **440** may extend from a distal end of the clevis shaft **434**. The ball socket ring **440** may be sized and configured to engage the probe holder **300** in such a manner that the probe holder **300** is capable of swiveling within the ball socket ring **440**, typically in a manner which will be hereinafter described. In some embodiments, a clevis attachment knob **438** may facilitate detachable attachment of the ball socket ring **440** to the clevis shaft **434** according to the knowledge of those skilled in the art.

(85) A ring tensioning knob **442** may engage the ball socket ring **440**. The ring tensioning knob **442** may be configured to threadably engage the ball socket ring **440** to facilitate selective adjustments in the width or diameter of the ball socket ring **440** for optimum engagement between the probe holder **300** and the ball socket ring **440**.

(86) As illustrated in FIGS. 20-34, the grip portion **302** of the probe holder **300** may include a grip portion wall **304** having a grip portion base **306**, a grip portion apex **308** and an expanded grip portion middle section **310**. A ball socket **370** may extend from the grip portion apex **308** of the grip portion **302**. The ball socket **370** may be suitably sized and configured to be received in the ball socket ring **440** of the holder mount clevis **432** for swivel mounting of the ball socket **370** in

the ball socket ring **440** and multi-positional mounting of the probe holder **300** with respect to the stand assembly **400**.

(87) The grip portion **302** may have a pair of mating grip portion sections **312**. The grip portion sections **312** of the grip portion **302** may be configured to detachably engage each other via a flanged connection, a tongue in groove connection, or the like. In some embodiments, the ball socket **370** may have a fixed ball socket section **372** and a detachable ball socket section **376**. The fixed ball socket section **372** may extend from one of the mating grip portion sections **312** of the grip portion **302**. The detachable ball socket section **376** may be configured to detachably engage the fixed ball socket section **372** at a suitable attachment interface **374**. Accordingly, in typical assembly of the probe holder **300** on the ultrasonic transducer probe **350**, as illustrated in FIG. 32 and will be hereinafter described, the grip portion sections **312** may detachably engage each other. The detachable ball socket section **376** may detachably engage the fixed ball socket section **372** to form the ball socket **370** for swivel attachment of the probe holder **300** to the stand assembly **400**.

(88) As further illustrated in FIG. 32, in some embodiments, each grip portion section **312** of the grip portion **302** may be fitted with a grip pad **324**. Each grip pad **324** may be inserted from the inside out through an elongated grip pad opening (not numbered) in each corresponding grip portion section **312**.

(89) As illustrated in FIG. 30, the skirt portion **326** may include a skirt portion wall **328** formed with a skirt portion base **330** and a skirt portion apex **332**. A central transducer mount portion **334** may be disposed in the skirt portion interior **342** of the skirt portion **326**. The transducer mount portion **334** may be suitably sized and configured to receive and support the probe base **356** of the ultrasonic transducer probe **350** in assembly of the grip portion **302** on the ultrasonic transducer probe **350**.

(90) The skirt portion wall **328** of the skirt portion **326** may be fabricated of a flexible material such as silicone, rubber or plastic, for example and without limitation. In some embodiments, the resiliency of the skirt portion wall **328** may render the skirt portion **326** capable of retracting or folding backwards, or turning inside out, in a manner which is similar to that of an umbrella being flipped inside out by a strong gust of wind, typically for purposes which will be hereinafter described. The flipped skirt portion **326** may be deployed back into position with a downward swipe of the hand.

(91) A vacuum tube port **394** and a gel tube port **396** may be provided in the skirt portion interior **342** of the skirt portion **326**. The vacuum tube end **384** of the vacuum tube **380** may terminate in fluid communication with the vacuum tube port **394**. The gel tube end **386** of the gel tube **382** may terminate in fluid communication with the gel tube port **396**. In some embodiments, the vacuum tube end **384** of the vacuum tube **380** and the gel tube end **386** of the gel tube **382** may be suitably sized and configured to insert into the respective vacuum tube port **394** and gel tube port **396**.

(92) The vacuum tube **380** and the gel tube **382** may extend from the skirt portion apex **332** of the skirt portion **326**. The vacuum tube **380** may be connected to a vacuum system **460** (FIG. 37), and the gel tube **382** may be connected to a pump and supply mechanism **450** (FIG. 37) which contains a conductive gel (not illustrated).

(93) As further illustrated in FIG. 30, in some embodiments, an annular vacuum distribution trough **388** may be provided in the skirt portion **326**. The vacuum distribution trough **388** may be disposed in fluid communication with the vacuum tube port **394**. The vacuum distribution trough **388** may be formed by and between the skirt portion wall **328** and the transducer mount portion **334** of the skirt portion **326**.

(94) As illustrated in FIGS. 35 and 36, a typical gel distribution circuit which is suitable for distributing gel from the pump and supply mechanism **450** to the skirt portion **326** of the probe holder **300** is shown. The gel tube **382** may establish fluid communication between the pump and supply mechanism **450** and the gel tube port **396** (FIG. 30) in the skirt portion interior **342** of the skirt portion **326**. At least one check valve **390** may be provided in the gel tube **382**. The check

valve **390** may facilitate unidirectional flow of the gel from the pump and supply mechanism **450** to the skirt portion **326** while preventing backflow of the gel through the gel tube **382**.

(95) As illustrated in FIG. **36**, in some embodiments, at least one pressure transducer **392** may be provided in the gel tube **382** of the gel distribution circuit. The pressure transducer **392** may be configured to measure the pressure of the gel as the gel flows through the gel tube **382**. In some embodiments, the pressure transducer **392** may operationally interface with the pump and supply mechanism **450** to facilitate feedback control of the pumping action for regulation of the pressure of the gel as the gel flows through the gel tube **382**. In some embodiments, a pressure gauge (not illustrated) may operationally interface with the pressure transducer **392**. The pressure gauge may be configured to visually indicate the pressure of the gel as the gel flows through the gel tube **382**.

(96) As illustrated in FIG. **37**, in some embodiments, a user interface **448** for at least some or all of the components of the medical ultrasound system may be provided on the stand base **402** of the stand assembly **400**. Accordingly, the user interface **448** may include a vacuum pressure control **454**. The vacuum pressure control **454** may operationally interface with the vacuum system **460**, typically through a processor **452**. The vacuum system **460** may be configured to control or regulate the vacuum pressure which is applied to the vacuum tube port **394** (FIG. **30**) through the vacuum tube **380** responsive to manipulation of the vacuum pressure control **454**. Accordingly, the vacuum pressure control **454** may be configured to control the pressure of the skirt portion **326** against the skin **364** of the patient **362** as the gel is dispensed from the skirt portion **326** of the probe holder **300** onto the skin **364** of the patient **362**. In some applications, for example and without limitation, the vacuum strength applied through the vacuum tube **380** may quickly ramp up and then level off or plateau at the preferred vacuum pressure strength.

(97) In some embodiments, the user interface **448** may include a gel quantity control **456**. The gel quantity control **456** may operationally interface with the pump and supply mechanism **450**, typically through the processor **452**. The pump and supply mechanism **450** may be configured to control or regulate the quantity or volume of the gel responsive to manipulation of the gel quantity control **456**. Accordingly, the gel quantity control **456** may thus be configured to control the quantity of volume of the gel dispensed from the skirt portion **326** of the probe holder **300** onto the skin **364** of the patient **362**.

(98) In some embodiments, the user interface **448** may include a gel temperature control **458**. The gel temperature control **458** may operationally interface with the pump and supply mechanism **450**. The pump and supply mechanism **450** may include one or more heating elements (not illustrated) configured to heat the gel responsive to manipulation of the gel temperature control **458**.

Additionally or alternatively, at least one warming coil (not illustrated) may be provided typically at or adjacent to the gel tube end **386** (FIG. **30**) of the gel tube **382** for warming of the gel before or as it is dispensed from the gel tube port **396** for increased comfort to the patient **362**. Accordingly, the gel temperature control **458** can be manipulated to control the temperature of the gel dispensed from the skirt portion **326** of the probe holder **300** onto the skin **364** of the patient **362**.

(99) The vacuum pressure control **454**, the gel quantity control **456** and/or the gel temperature control **458** and/or other features of the user interface **448** may be provided in any location or position which is accessible to the ultrasound technician **398**. For example and without limitation, in some embodiments, the user interface **448** may be provided on the main base member **404** of the stand base **402** of the stand assembly **400**. Additional controls of the user interface **448**, such as vacuum power, sustain, and/or time settings, for example and without limitation, may be included on the main base member **404** of the stand base **402** and/or in any other suitable accessible location or locations. In some embodiments, the vacuum pressure control **454**, the gel quantity control **456**, the gel temperature control **458** and/or other features of the user interface **448** may include a touch sensor with circuitry known by those skilled in the art.

(100) As illustrated in FIGS. **38-52**, in typical application, the stand assembly **400** may be mounted on the support **348** on which the patient **362** reclines. Accordingly, the ultrasound technician **398**

may adjust the spacing between the base member flange **406** and the main base member **404** of the stand base **402**, typically by rotation of the base adjustment knob **418**, depending on the thickness of the rail, panel, or other element on the support **348** to which the stand assembly **400** is mounted. (101) As illustrated in FIG. **38**, the support **348** may be placed between the main base member **404** and the base member flange **406**. As illustrated in FIG. **39**, the base adjustment knob **418** may then be rotated to clamp or tighten the support **348** between the base member flange **406** and the main base member **404**.

(102) As illustrated in FIG. **40**, the ultrasonic transducer probe **350** may next be fed through the ball socket ring **440** of the holder mount clevis **432** on the stand assembly **400** until the probe cable **358** extends through the ball socket ring **440**. The ultrasonic transducer probe **350** may then be placed inside the grip portion section **312** of the grip portion **302** from which the fixed ball socket section **372** of the ball socket **370** extends, as illustrated in FIG. **41**. As illustrated in FIG. **42**, the detachable ball socket section **376** may then be snapped in place on the fixed ball socket section **372** to complete assembly of the ball socket **370**.

(103) As illustrated in FIG. **43**, the remaining grip portion section **312** may be snapped in place on the first grip portion section **312** to complete assembly of the grip portion **302**. The remaining grip portion section **312** may remain structurally separate from the ball socket **370** to facilitate cleaning of the ultrasonic transducer probe **350** without the need to completely remove the ultrasonic transducer probe **350** from the ball socket **370**. As illustrated in FIG. **44**, the ball socket **370** may then be inserted into the ball socket ring **440** of the holder mount clevis **432**.

(104) As illustrated in FIG. **45**, the flexible skirt portion **326** may be placed over the probe base **356** of the ultrasonic transducer probe **350** as the vacuum tube end **384** (FIG. **30**) of the vacuum tube **380** is inserted into the vacuum tube port **394**. The gel tube end **386** of the gel tube **382** may be inserted into the gel tube port **396**. The ball socket ring **440** of the holder mount clevis **432** may next be tightened against the ball socket **370** of the probe holder **300**, typically by rotating the ring tensioning knob **442**, as illustrated in FIG. **46**.

(105) As illustrated in FIG. **47**, the medical ultrasound system may be powered on typically by manipulation of the user interface **448**. Settings such as vacuum power, sustain, and/or time settings may be adjusted by the ultrasonic technician **398**. For example and without limitation, as illustrated in FIG. **37**, in some applications, the vacuum pressure control **454**, the gel quantity control **456** and/or the gel temperature control **458** may be manipulated by the ultrasonic technician **398** to facilitate adjustments in the vacuum pressure, the gel quantity or volume and/or the gel temperature, respectively.

(106) As illustrated in FIG. **48**, the ultrasound technician **398** may grasp the probe holder **300** and position the skirt portion **326** against the skin **364** of the patient **362**. The position of the probe holder **300** may be manipulated until the desired view on the patient **362** is found. The ultrasonic transducer probe **350** may then be activated by squeezing the grip pads **324** on the respective grip portion sections **312** of the grip portion **302**, typically twice. This action may cause the vacuum system to apply vacuum pressure to the vacuum tube port **394** (FIG. **30**) through the vacuum tube **380**, thereby applying vacuum pressure to the annular vacuum distribution trough **388** in the skirt portion **326**. Accordingly, uniform vacuum pressure in the skirt portion interior **342** of the skirt portion **326** may facilitate application of the skirt portion **326** to the skin **364** of the patient **362** along a uniform seal, gently holding the probe base **356** of the ultrasonic transducer probe **350** against the patient's skin **364**, as illustrated in FIG. **49**. The ultrasound technician **398** is thus free to view the ultrasound data which is outputted by the ultrasonic transducer probe **350** without having to sustain manual pressure of the ultrasonic transducer probe **350** against the patient's skin **364**, as illustrated in FIG. **50**.

(107) As illustrated in FIG. **51**, after the ultrasonic viewing of the patient is completed, the ultrasound technician **398** may terminate further application of vacuum pressure to the vacuum tube port **394** typically by again squeezing the grip pads **324** on the grip portion **302**. As illustrated

in FIG. 52, the probe holder **300** with the ultrasonic transducer probe **350** therein may then be moved to a different location on the skin **364** of the patient **362** to obtain another image, typically in the manner which was heretofore described.

(108) In some applications, the ultrasonic transducer probe **350** may be used to obtain a view known as a supra sternal notch view which is taken with the probe **350** applied just beneath the “Adams apple” of the patient **362**. This view may require direct exposure of the probe base **356** of the probe **350**. As set forth herein above, to achieve this purpose, the skirt portion **326** may be designed to retract or fold backwards with a simple swipe of the hand. The skirt portion **326** may subsequently be deployed back into position with a downward swipe of the hand.

(109) It will be appreciated by those skilled in the art that the probe holders **100**, **300** represent an immediate and universally applicable method to markedly lessen the musculoskeletal traumas associated with chronic sonography activities. Its simple design and ease of application is the first of its kind to reduce musculoskeletal traumas most directly and efficiently during the acquisition of images. It is a relatively low-cost method that allows the sonographer and institutions to use their current probes, protocol, and computer machinery as they are already accustomed. The lamprey acts like an invisible assistant sonographer that unburdens the actual sonographer from the daily brunt force work allowing them to maintain proper posture and markedly reduce the possibility of musculoskeletal trauma thereby avoiding career ending muscle and joint strain. Conventionally, sonographers identify a particular probe position which is maintained with one hand with application of pressure while the free hand does typing, measurements and manipulations with the computer keyboard. Another location is then identified, and new sets of pictures obtained. An average to complex study may have 60 to 120 images. Normally the dominant hand using the probe maintains pressure and positioning throughout the length of the study. Now with introduction of the new lamprey accessory the sonographer only has to locate the line of sight of their image and release the probe freeing the hand, wrist, arm and shoulder from any further use of pressure. Upon release the probe position in space is maintained and vacuum pressure is initiated producing a downward force of the probe into the chest wall as would normally be provided by the sonographer. The position is held as long as needed while the sonographer makes additional measurements and recordings before proceeding to the next imaging location. Upon touching and repositioning the probe the vacuum releases and upon securing the next position the vacuum engages when the hand is released from the probe. The probe maintains neutral buoyancy in space allowing for easy maneuverability. No forceful tugging of the probe is required further minimizing any muscle strain. The lamprey will allow future echocardiographic studies with negligible inherent muscular trauma and provide built-in muscle recovery time. The lamprey should significantly change the field of sonography allowing more senior sonographers to continue imaging later into their careers adding to the knowledge base and proficiency of echocardiography programs.

(110) Since many modifications, variations, and changes in detail can be made to the described preferred embodiments of the invention, it is intended that all matters in the foregoing description and shown in the accompanying drawings be interpreted as illustrative and not in a limiting sense. Thus, the scope of the invention should be determined by the appended claims and their legal equivalents.

Claims

1. An ultrasonic transducer probe holder suitable for holding an ultrasonic transducer probe of a medical ultrasound system in place against the skin of a patient as an ultrasound technician manually releases the probe, the medical ultrasound system having a pump and supply mechanism configured for pumping conductive gel and a vacuum system configured for applying vacuum pressure to the ultrasonic transducer probe holder, the ultrasonic transducer probe holder

comprising: a grip portion configured for deployment on the ultrasonic transducer probe, the grip portion comprising a flexible grip portion wall including: a grip portion base; a grip portion apex; a grip portion middle section between the grip portion base and the grip portion apex, the grip portion middle section of the grip portion wall expandable to a greater diameter or width than a diameter or width of the grip portion base and the grip portion apex of the grip portion wall; and the grip portion comprising a pair of detachable, mating grip portion sections; a skirt portion configured to form a vacuum seal against the skin of the patient responsive to operation of the vacuum system, the skirt portion further configured to support the grip portion and comprising: a flexible skirt portion wall having a skirt portion apex at the grip portion; a skirt portion base opposite the skirt portion apex; a skirt portion interior formed by the skirt portion wall; and a central transducer mount portion disposed in the skirt portion interior; a ball socket on the grip portion, the ball socket comprising: a fixed ball socket section extending from a first one of the mating grip portion sections of the grip portion; and a detachable ball socket section attachable to the fixed ball socket section and to a second one of the mating grip portion sections of the grip portion; and a stand assembly configured for mounting on a support, the stand assembly comprising: a stand base; at least one stand arm pivotally supported by the stand base; and a holder mount clevis pivotally supported by the at least one stand arm, the holder mount clevis comprising a ball socket ring, wherein the ball socket on the grip portion is insertable in the ball socket ring.

2. The ultrasonic transducer probe holder of claim 1 further comprising a vacuum tube port in the skirt portion interior of the skirt portion, and wherein the vacuum tube is configured for placement into fluid communication with the vacuum tube system.

3. The ultrasonic transducer probe holder of claim 1 further comprising a gel tube configured for placement into fluid communication with the skirt portion interior, the gel tube configured to distribute conductive gel to the skirt portion interior of the skirt portion.

4. The ultrasonic transducer probe holder of claim 1 wherein the at least one stand arm comprises a first stand arm pivotally supported by the stand base and a second stand arm pivotally supported by the first stand arm, and wherein the holder mount clevis is pivotally supported by the second stand arm.

5. The ultrasonic transducer probe holder of claim 1 further comprising a user interface on the stand base of the stand assembly, the user interface configured to facilitate operation of the pump and supply mechanism and the vacuum system of the medical ultrasound system.

6. The ultrasonic transducer probe holder of claim 5 wherein the user interface comprises a vacuum pressure control, and gel quantity control and a gel temperature control.
